

Oracle Database Link Practical Guide

TECHNICAL DOCUMENTATION & IMPLEMENTATION REFERENCE

1. Public Database Link

Definition: Available to all users and schemas within the local database instance.

Creation Script

```
CREATE PUBLIC DATABASE LINK db_orcl CONNECT TO hr IDENTIFIED BY hr_password USING
'orcl';
```

Database link created.

Verification Check

```
SELECT OWNER, DB_LINK, USERNAME, VALID FROM DBA_DB_LINKS;
```

OWNER	DB_LINK	USERNAME	VAL
SYS	SYS_HUB		YES
PUBLIC	DB_ORCL	HR	YES

Testing Execution via Another Schema (e.g., SCOTT)

```
-- Connected as scott/tiger SELECT * FROM locations@db_orcl;
```

```
LOCATION_ID STREET_ADDRESS POSTAL_CODE CITY STATE_PROVINCE CO -----
----- -- 1000 1297 Via Cola di Rie 00898 Roma IT
1100 93091 Calle della Testa 10934 Venice IT 1200 2017 Shinjuku-ku 1689 Tokyo Tokyo Prefect. JP
1300 9450 Kamiya-cho 6823 Hiroshima JP 1400 2014 Jabberwocky Rd 26192 Southlake Texas US
```

Drop Syntax

```
DROP PUBLIC DATABASE LINK db_orcl;
```

Database link dropped.

2. Private Database Link

Definition: Restricted for use only by a specific schema user within the local database.

Privilege Grant & Creation

```
-- Granted by DBA GRANT CREATE DATABASE LINK TO scott; -- Created within SCOTT schema
CONNECT scott/tiger; CREATE DATABASE LINK DB_SCOTT CONNECT TO SCOTT IDENTIFIED BY TIGER USING
'ORCL';
```

Database link created.

Global Registry Verification (DBA view)

```
SELECT OWNER, DB_LINK, USERNAME, VALID FROM DBA_DB_LINKS;
```

OWNER	DB_LINK	USERNAME	VAL
SYS	SYS_HUB		YES
PUBLIC	DB_ORCL	HR	YES
SCOTT	DB_SCOTT	SCOTT	YES

Query Execution

```
SHOW USER; -- USER is "SCOTT"
SELECT * FROM emp@db_scott;
```

```
ID NAME LOCATION DEPARTMENT -----
-- 1 Anup Kolkata IT 2 Alok Kolkata Account 3 Brishali Kolkata HR
```

3. Summary: Public vs. Private DB Links

Feature	Private DB Link	Public DB Link
Owner	Specific Schema User (e.g., SCOTT)	PUBLIC
Accessibility	Owner schema only	All users in the local database
Required Privilege	CREATE DATABASE LINK	CREATE PUBLIC DATABASE LINK
Drop Syntax	DROP DATABASE LINK link_name;	DROP PUBLIC DATABASE LINK link_name;

Best Practice Note: Always verify remote network descriptors in tnsnames .ora prior to link creation, and use dedicated accounts with minimum required access permissions.